

flea

NEWS 89 *March 2022*



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FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Before including in publications any of the citations given here, please review for accuracy and correct formatting. Many of our sources are abstracting journals and current literature sources, and editors have not necessarily checked citations.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: New flea species descriptions, Flea research citations from journals or conference proceedings that are not available or indexed, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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Assistant Editor J. R. Kucera, M.S.

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Editorial

“*What we do not know about fleas*”

Dear *Flea News* Reader,

I have often been bemused by my knowledge of natural science: It is “a mile wide, and an inch deep”. I know very much about a very little, specialization compounding my deficiency.

Stephen Hawking (2018) remarked:

“In the 18th century, there was said to be a man who had read every book written. But nowadays... it would take you many tens of thousands of years... no one person can be the master of more than a small corner of human knowledge. People have to specialize, in narrower and narrower fields. This is likely to be a major limitation in the future. We certainly cannot continue, for long, with the exponential rate of growth of knowledge that we have had in the last 300 years.”

A shallow understanding is spot-on of entomology, where we lack information on even basic biology of many insects.

In the last several *Flea News* issues, I have argued for research focusing on the importance of parasites, especially fleas, and their microbial symbiotes, as keystone species and ecological engineers within ecosystems. I noted that vertebrate biologists and even parasitologists often seem to neglect the influence of parasites in maintaining diversity, and assuring homeostasis of ecosystem functions.

Yet, it is possible to take broad views, as Grimaldi (2000) who listed insects that are keys for the evolving of Earth’s biosphere:

Termites

Hemipterans

Phytophagous Beetles

Cyclorrhaphan Flies

Glossatan Moths

Hymenoptera

Hölldobler and Wilson (2009) pointed out Flowering Plants, Nematodes, Spiders, and Mites, too, are noteworthy, as non-insect accessories often closely tied to insects, in "creating ever more complex ecosystems". Although almost all of these groups contain parasitic species, the well-known parasite groups including fleas are conspicuous by their absence.

Hölldobler and Wilson emphasize how much we do not yet understand. Even for the better-studied ants, their:

"...unfolding, however, can still be seen only in fragments. Large gaps remain in the fossil record, especially across the critical period of major radiation that reached from the Late Cretaceous into the Paleogene. Of equal consequence, the life cycles and natural history of the vast majority of living species, which still bear the indelible stamp of this history, remain unexplored".

We may say the same of fleas; these gaps are *terra incognita* in our considerations. Holland's (1964) summary remains valid:

"... we are a long way from understanding the ecology and physiology of fleas, and from tracing a meaningful history of their host associations."

Biologists have literally centuries of work before them.

Continued best wishes concerning the coronavirus pandemic.

Yours in fleas,
R.L. Bossard
Editor, *Flea News*

Literature

Grimaldi, D. 2000. Mesozoic radiations of the insects and origins of the modern fauna. Proc. 21st Intl. Congress of Entomology 1: xix-xxvii.

Hölldobler, B. and Wilson, E.O. 2009. The superorganism. W.W. Norton Publ., New York, pg. 315, 330.

Holland, G.P. 1964. Evolution, classification, and host relationships of Siphonaptera. Annu. Rev. Entomol. 1964.9:123-146.

Hawking, S. 2018. Brief answers to the big questions. Penguin Random House, New York.

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Announcements

The Livestock Insect Worker's Conference 2022
June 6-8th
will be returning to an in-person meeting in Dallas, Texas.

Flea Presentations at the Entomological Society of America annual meeting 2021

Denver Colorado Oct. 31- Nov. 3, 2021

Options for flea and tick control

Nancy C. Hinkle, Jacob W. Fowler

University of Georgia, Athens, Georgia

Expanding ranges of many tick species in North America expose more animals, including pets, to ticks and create concern among pet owners. Options for tick and flea suppression are meager, with only a few modes of action represented among active ingredients available. Treatment methods (spray, dust, spot-on, oral, etc.) are likewise limited.

Spatial variation in fleas and flea-borne pathogens collected from urban and sylvatic opossums and raccoons in north-central Florida

Sarah Mays Maestas, Lindsay P. Campbell, Samantha M. Wisely, Lawrence Reeves, Phillip E. Kaufman

University of Florida, Vero Beach, Florida; Gainesville, Florida; Texas A&M University, College Station, Texas

Mammalian wildlife species are hosts of ectoparasite species that also affect humans and companion animals, and which may be associated with the transmission of vector-borne zoonotic disease agents. Increasing urbanization and the adaptation of wildlife to urban environments may result in changes in the ecology of ectoparasites and vector-borne diseases in urban areas compared to sylvatic areas. Fleas collected from live-trapped opossums and raccoons from a rural, wooded habitat, and from the city of Gainesville, FL over four seasons were screened for *Bartonella* and *Rickettsia* infection. Redundancy analyses revealed differences in flea community composition across host species and sylvatic vs. urban habitat, with *Ctenocephalides felis* being more closely associated with urban opossums and *Xenopsylla cheopis* being more closely associated with sylvatic opossums. *Orchopeas howardi* were collected on both opossums and raccoons in urban and sylvatic sites. Of the 1,017 fleas screened for pathogens, 383 were infected with *Rickettsia* species. There was an ~40% *Rickettsia* prevalence in fleas collected from urban animals, and an ~9% *Rickettsia* prevalence in fleas collected from sylvatic animals. Of the samples currently sequenced, 78% were infected with *Rickettsia felis*. Eight fleas were infected with *Bartonella* species. The high prevalence of *Rickettsia* in fleas from urban animals may have implications for human, companion animal, and wildlife health in urban settings. We will continue to explore how season and landscape factors may be associated with pathogen prevalence in urban and sylvatic habitat to contribute to improved vector and pathogen surveillance efforts.

Temperature phenology of fleas (Siphonaptera) on rodents (Mammalia: Rodentia)

Robert L. Bossard

(abstract in Flea News 88)

Tick- and flea-borne disease surveillance along the Texas/Mexico border

Christopher J. Vitek, Valerie Hernandez, Conseulo Aguilar, Nicole Mendell, Donald Bouyer

University of Texas Rio Grande Valley, Edinburg, Texas; McAllen, Texas; University of Texas Medical Branch, Galveston, Texas, United States

Tick- and flea-borne diseases in South Texas have historically not received significant attention due to the high temperatures and lack of reported diseases. However, recent incidences of flea-borne typhus and rickettsial diseases in Texas have highlighted the need for increased understanding of tick- and flea-borne diseases. Working in conjunction with other partners in the Western Gulf Center of Excellence in Vector-Borne Diseases, we initiated and participated in a long-term collection, identification, and testing effort for ticks and fleas to assess the risk of common tick- and flea-borne diseases. While the 2020 season was disrupted in March 2020 with SARS-COV2 concerns, surveillance activities have resumed in September 2020. Collection sites have included local animal shelters, field collections, and partnering with veterinarians to receive samples. To date, we have identified 2132 ticks, consisting of 4 species and 344 fleas, consisting of 4 species. The most common tick species collected in animal shelters and veterinarian clinics was *Rhipicephalus sanguineus*, and the most common flea species was *Ctenocephalides felis* [sic, *Ctenocephalides felis*]. Testing has identified *Rickettsia amblyommatis*, *R. parkeri*, and *Ehrlichia chaffeensis* [sic, *chaffeensis*] in the ticks and is ongoing within the fleas. We discuss the temporal and geographic patterns, as well as future efforts to expand this study.

Changing *Peromyscus* distributions in the upper Midwest and consequences for tick-borne pathogens

Ryan T. Larson, Gebbiena Bron, Xia Lee, Tela Zembsch, Scott Larson, Susan M. Paskewitz

US Naval Medical Research Unit No. 6, Dulles, Virginia; Wageningen University, Wageningen, Gelderland, Netherlands,; University of Wisconsin Madison; Metropolitan Mosquito Control District.

In the United States, mice are the most abundant small mammal reservoir of the Lyme disease spirochete and consequently are responsible for infecting a high proportion of blacklegged ticks, *Ixodes scapularis* Say. Long-term observations in the Midwest suggest warming temperatures are promoting poleward expansion of the white-footed mouse, *Peromyscus leucopus* Rafinesque, as well as replacement of the woodland deer mouse, *Peromyscus maniculatus gracilis* Wagner. We used species

distribution modeling (SDM) and joint species distribution modeling (JSDM) approaches to identify the role of abiotic and biotic factors in shaping the distributions of two sympatric tick-borne pathogen reservoirs, *Peromyscus maniculatus* and *P. leucopus*, in Minnesota and Wisconsin. Additionally, we predicted the future distributions for *P. leucopus* based on climate projections. Finally, we conducted small mammal trapping in north-central Wisconsin, where both species occur, to compare tick infestations and *Borrelia burgdorferi* prevalence. We found no evidence that these *Peromyscus* species co-occurred less than expected given their shared responses to environmental variables. The number of frost-free days was the most important variable describing distributions for both mice and future projections suggest that the northernmost areas of Wisconsin and Minnesota will become more suitable for *P. leucopus*. Although *P. maniculatus* were more likely to be infected with *Borrelia burgdorferi* compared to its congener, *P. leucopus* were more infested with both larval and nymphal *I. scapularis*. Collectively, results suggest that shorter-milder winters could promote further expansion of *P. leucopus*, a more competent reservoir of tick-borne pathogens, in the Midwest.

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A macroevolutionary view of host preference in mosquitoes

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The necessity of many mosquito species to feed from the blood of vertebrate hosts for egg production, thus exposing host and vector to potential pathogens, has caused significant interest in the evolution of host preference. Most of these analyses have focused on the evolution of host preference within a single population or species. Here, utilizing the most complete phylogenetic analysis of mosquitoes to date, we evaluate the evolution of blood host preference in mosquitoes in a macroevolutionary framework, leveraging phylogenetic comparative methods to describe the evolution of blood host preference across 180 million years of mosquito evolution.

M.S. Thesis

MELLIS, ANNA MARIE (2021) Spatial Variation in Mammal and Ectoparasite Communities in the Foothills along the Southern Appalachian Mountains. M.S. thesis, University of North Carolina at Greensboro.

Small mammal and ectoparasite community variation and abundance is important for monitoring the transmission rate of zoonotic diseases and informing conservation efforts that maintain host and parasite biodiversity in ecosystems facing global climate change. The purpose of this study was to identify the factors driving variation in small mammal and ectoparasite communities in the Southern Appalachian Mountains. I took an approach to sampling that allowed me to test predictions

from island biogeography theory; namely, that host species richness varies with distance from the main Appalachian mountain range. I also examined how ectoparasite species richness varied with small mammal richness as well as ecological variables. Finally, I analyzed ectoparasite abundances at the community- and individual-host levels to understand how changes in host species richness may affect infestation rates. Comprehensive field surveys and ectoparasite screenings were performed across four field sites, two isolated from the Southern Appalachian Mountains and two along the Southern Appalachian Mountains. I found that these field sites were characterized by a mix of high and low elevation mammal species, and that community structure varied with degree of isolation for mammals, but not ectoparasites. Habitat type was a significant driver of species variation within and among sites. I found decreased abundances in ectoparasite compound communities when host species diversity was highest, which is consistent with predictions from a dilution effect. However, when evaluating abundances of individual ectoparasites, only one (*Leptotrombidium peromysci*) of four species displayed patterns consistent a dilution effect. My results provide new information on small mammal distributions and ectoparasite associations at disparate sites across the foothills of the Southern Appalachians but suggest that host-parasite associations and the intensity of infection are subject to additional environmental or ecological drivers beyond those investigated here.

[Editor's note: fleas were *Polygenis gwyni* (Pg), *Ctenophthalmus pseudagyrtis* (Cp), *Epitedia wenmanni* (Ew), *Doratopsylla blarinae* (Db), *Peromyscopsylla hesperomys* (Ph), and *Orchopeas leucopus* (Ol); also found Phthiraptera, Ixodida, Acariformes, Trombidiformes.]

Featured Research

New Flea Species 2021

Guerra-Murcia, N., Acosta, R., Gatica-Colima, A., Vital-García, C., & Rueda-Torres, R. (2021). New species of flea of the genus *Dactylopsylla* Jordan (Insecta: Siphonaptera) from the Flora and Fauna Protection Area Médanos de Samalayuca, Chihuahua, Mexico. *Zootaxa*, 5032(1), 113-120. doi:10.11646/zootaxa.5032.1.6

A new species of *Dactylopsylla* Jordan, parasites of Geomyidae rodents from Chihuahua, is described and illustrated as *D. samalayuca* n. sp. This species is compared with their morphologically closest relatives. Males are characterized by the shape of the upper lobe of the immovable [sic, immovable] and movable process, the shape and chaetotaxy of the distal arm of sternum IX and the shape of the crochet; and females by the contour of the distal margin of sternum VII. The geographical distribution of *Dactylopsylla* is extended to the Chihuahuan desert in Mexico as *D. samalayuca* n. sp. is reported from south of the Natural Protected Area Mdanos de Samalayuca (MS). A recent key to *Dactylopsylla* species is updated with inclusion of the new species.

[*Dactylopsylla samalayuca* Guerra-Murcia, et al. 2021]

GONG Zheng-da [gongzd@126.com], XU Xiang, LI Chun-fu, WEN Xiang-bing, ZI Jin-rong. 2021. A study of the taxonomy and fauna of Siphonaptera in Tibet autonomous region, China. Description of two new species (Siphonaptera: Ctenophthalmidae) from Southeast Tibet, China. *Chin J Vector Biol &*

Control, 2021, 32(1): 85-88. doi: 10.11853/j.issn.1003.8280.2021.01.018

In this article, two new species from genera *Xenodaeria* Jordan (1932) and *Rhadinopsylla* Jordan et Rothschild (1912) (Siphonaptera: Ctenophthalmidae), collected from Linzhi, Tibet autonomous region of China, were described. (1) *Xenodaeria houyongi* sp. nov. Its main features are as follows: The frontal bristle consists of six spiniform bristles with deep ossification; in females, the depression of the posterior margin of sternum VII was small and shallow, the ventral lobe was wide and large and significantly longer than the dorsal lobe, and the height of the dorsal margin is significantly lower than that of the 7th spiracle; the depression of ventral margin at the bulga of spermatheca is located at the midpoint, and the hilla end has papillae. The holotype (only one ♀) was collected from the body of *Crocidura dracula* in a farming area (at an altitude of about 1 200 m) of Motuo county, Linzhi in December 2019. (2) *Rhadinopsylla (Actenophthalmus) linzhiensis* sp. nov. Its main features are as follows: It has a total of 18 pronotal combs [see editor's note below]; in males, the movable process of clasper is straight and slender, with symmetrical front and rear margins; the sternum VIII has only 2 bristles; the apex port of posterior arms at sternum IX has a series of spiniform bristles; the end of uncinat process of the aedeagus is finger-like. The holotype (only one ♂) was collected from the body of *Ochtona* sp. in a farming area (at an altitude of about 3 000 m) of Bayi town, Linzhi in November 2018.

[*Xenodaeria houyongi* Gong, Xu & Zi 2021]

[*Rhadinopsylla (Actenophthalmus) linzhiensis* Gong, Li & Wen 2021]

[Editor's note: Presumably, the authors are saying that there are 18 spines in the pronotal comb.]

Hastriter, M. W. (2021). Fleas (Siphonaptera: Pygiopsyllomorpha) of Papua New Guinea and Papua province (Indonesia). Part VI. *Bibikovana*, *Geohollandia*, and *Hoogstraalia* (Pygiopsyllidae: Pygiopsyllinae), with descriptions of four new species. *Annals of Carnegie Museum*, 87(1), 37-77. doi:10.2992/007.087.0102

[Abstract in Flea News 88]

[*Bibikovana engilisi* Hastriter, 2021, *Bibikovana bosaviensis* Hastriter, 2021, *Bibikovana mekiliensis* Hastriter, 2021, and *Hoogstraalia nadchatrami* Hastriter, 2021]

Hastriter, M. W. (2021). Records of Fleas (Siphonaptera) from Australia, Malaysia, and Papua New Guinea with the Description of a New Species of *Bibikovana* Traub, 1980 (Pygiopsyllidae). *Annals of Carnegie Museum*, 87(2), 117-137.

[Abstract in Flea News 88]

[*Bibikovana acumena* Hastriter, 2021]

Keskin, A. (2021). Description of a new flea species of the genus *Ctenophthalmus* Kolenati (Siphonaptera: Ctenophthalmidae) from Turkey. *Parasitology Research*, 120(4), 1203-1207.

[Abstract in Flea News 88]

[*Ctenophthalmus (Ctenophthalmus) yesarii* Keskin, 2021]

Keskin, A. (2020). A new flea species of the genus *Palaeopsylla* (Insecta: Siphonaptera: Ctenophthalmidae) from Turkey. *Journal of medical entomology*, 57(1), 88-91.

[Abstract in Flea News 88]

[*Palaeopsylla (Palaeopsylla) beaucournui* Keskin, 2020]

Taxonomy

Cull, B. (2021) Potential for online crowdsourced biological recording data to complement surveillance for arthropod vectors. *PLoS ONE*, 16 (4 April), art. no. e0250382. DOI: 10.1371/journal.pone.0250382

Voluntary contributions by citizen scientists can gather large datasets covering wide geographical areas, and are increasingly utilized by researchers for multiple applications, including arthropod vector surveillance. Online platforms such as iNaturalist accumulate crowdsourced biological observations from around the world and these data could also be useful for monitoring vectors. The aim of this study was to explore the availability of observations of important vector taxa on the iNaturalist platform and examine the utility of these data to complement existing vector surveillance activities. Of ten vector taxa investigated, records were most numerous for mosquitoes (Culicidae; 23,018 records, 222 species) and ticks (Ixodida; 16,214 records, 87 species), with most data from 2019-2020. Case studies were performed to assess whether images associated with records were of sufficient quality to identify species and compare iNaturalist observations of vector species to the known situation at the state, national and regional level based on existing published data. Firstly, tick data collected at the national (United Kingdom) or state (Minnesota, USA) level were sufficient to determine seasonal occurrence and distribution patterns of important tick species, and were able to corroborate and complement known trends in tick distribution. Importantly, tick species with expanding distributions (*Haemaphysalis punctata* in the UK, and *Amblyomma americanum* in Minnesota) were also detected. Secondly, using iNaturalist data to monitor expanding tick species in Europe (*Hyalomma* spp.) and the USA (*Haemaphysalis longicornis*), and invasive *Aedes* mosquitoes in Europe, showed potential for tracking these species within their known range as well as identifying possible areas of expansion. Despite known limitations associated with crowdsourced data, this study shows that iNaturalist can be a valuable source of information on vector distribution and seasonality that could be used to supplement existing vector surveillance data, especially at a time when many surveillance programs may have been interrupted by COVID-19 restrictions.

[Editor's comment: Is there any use of iNaturalist for common flea distributions?]

Samuel, P.P., Govindarajan, R., Krishnamoorthi, R., Venkatesh, A. (2021) A rapid protocol for clearing, staining, and mounting of Arthropoda: Trombiculidae, Pediculidae and Pulicidae. North-Western

Journal of Zoology, 17 (1), pp. 1-5.

This is a rapid protocol proposed after its prolonged usage for the clearing, staining, and mounting of the medically important hematophagous ectoparasites obtained from human and small mammals collected in Tamil Nadu and Kerala, India. During the biodiversity study on ectoparasites, this altered protocol improved significantly the visibility of the major essential taxonomic key characters present on the minute body of the uncharacterized chigger mites, fleas, and lice viewed under phase-contrast microscopy for identification. This modified method will be enormously useful for imparting training, teaching, and research on acarology.

Sevestre, J., Diarra, A.Z., Laroche, M., Almeras, L., Parola, P. (2021) Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry: An emerging tool for studying the vectors of human infectious diseases. *Future Microbiology*, 16 (5), pp. 323-340. DOI: 10.2217/fmb-2020-0145

Arthropod vectors have historically been identified morphologically, and more recently using molecular biology methods. However, both of these methods are time-consuming and require specific expertise and equipment. Matrix-assisted laser desorption ionization time-of-flight mass spectrometry, which has revolutionized the routine identification of microorganisms in clinical microbiology laboratories, was recently successfully applied to the identification of arthropod vectors. Since then, the robustness of this identification technique has been confirmed, extended to a large panel of arthropod vectors, and assessed for detecting blood feeding behavior and identifying the infection status in regard to certain pathogenic agents. In this study, we summarize the state-of-the-art of matrix-assisted laser desorption ionization time-of-flight mass spectrometry applied to the identification of arthropod vectors (ticks, mosquitoes, phlebotomine sand-flies, fleas, triatomines, lice and Culicoides), their trophic preferences and their ability to discriminate between infection statuses.

[Editor's comment: The main advantage is that, although the equipment for matrix-assisted laser desorption ionization time-of-flight mass spectrometry (MALDI-TOF) is expensive, processing is pennies per test.]

van der Mescht, L., Matthee, S., & Matthee, C. A. (2021). New taxonomic and evolutionary insights relevant to the cat flea, *Ctenocephalides felis*: A geographic perspective. *Molecular Phylogenetics and Evolution*, 155 doi:10.1016/j.ympev.2020.106990

Hosts and Distribution

Fantozzi, M. C., Sanchez, J.P., Lareschi, M., Beldomenico P. M. 2021. Effects of host factors on the dynamics of fleas (Siphonaptera) in Sigmodontinae rodents (Cricetidae) from El Espinal Ecoregion, Argentina. *Acta Tropica* 225: 106177. doi: 10.1016/j.actatropica.2021.106177

Ezquiaga M. C.; Gallo, J.; D' Agostino, R.; Udrizar Sauthier, D.; Abba A. M.; Sanchez, J.P. 2021. Fleas and ticks in armadillos from Argentinean Patagonia: Diversity, abundance and distribution. *Acta*

Tropica, 219 (2): 105911. doi: 10.1016/j.actatropica.2021.105911

Lopez Berrizbeitia, M.F.; Hastriter, M.; Sanchez, J.P.; Diaz M. 2021. New record of *Alectopsylla unisetosa* (Siphonaptera: Ischnopsyllidae) from Patagonia. Parasitology International, 85:102426. doi: 10.1016/j.parint.2021.102426

Ruiz, M.; Acosta, D.; Baricalla, A.; Sanchez, J.P. 2021. Molecular detection of *Rickettsia* in ectoparasites (Siphonaptera and Phthiraptera) of domestic and feral pigs from Argentina. Parasitology Research 120(3). doi:10.1007/s00436-021-07291-9

Galloway, Terry D. 2020. Lice (Phthiraptera: Trichodectidae), fleas (Siphonaptera: Pulicidae, Ceratophyllidae) and ticks (Ixodida: Ixodidae) infesting American badger, *Taxidea taxus* (Mammalia: Mustelidae), in Manitoba, Canada. Proceedings of the Entomological Society of Manitoba, Volume 76: 27-36.

[*Pulex irritans* Linnaeus, *Oropsylla rupestris* (Jordan), *O. bruneri* (Baker), *O. tuberculata* (Baker), and *O. arctomys* (Baker)]

Ilinsky, Y.; Lapshina, V.; Verzhutsky, D.; Fedorova, Y.; Medvedev, S. 2022. Genetic Evidence of an Isolation Barrier between Flea Subspecies of *Citellophilus tesquorum* (Wagner, 1898) (Siphonaptera: Ceratophyllidae). Insects 2022, 13, 126. <https://doi.org/10.3390/insects13020126>

We studied field collections of two flea subspecies that are plague vectors on the vast Palearctic territory. Analysing the molecular–genetic, geographical, morphological, and reproductive isolation criteria, we conclude that these subspecies could be considered different species.

This study investigated the relationship between two subspecies of the *Citellophilus tesquorum* flea, *C. t. altaicus* and *C. t. sungaris*, which are vectors of the bacterium *Yersinia pestis* that causes human plague across the vast territories of the Palearctic. Adult fleas were collected from 16 localities and 11 populations in 2019 and 2020. Specimens were morphologically verified for subspecies status and analysed for mitochondrial cytochrome c oxidase subunit I (COI) DNA, nuclear ribosomal cluster internal transcribed spacer 1 (ITS1) and ITS2, and *Wolbachia*-infection status. Our results demonstrated a genetic difference between *C. t. altaicus* and *C. t. sungaris*. According to mitochondrial data, the genetic distance between clades of *C. t. altaicus* and *C. t. sungaris* was comparable with the species divergence of the genus *Callopsylla*, which is closely related to *Citellophilus*. All studied populations of *C. t. altaicus* were *Wolbachia*-infected, whereas all studied populations of *C. t. sungaris* were symbiont-free. Data for ITS1 and ITS2 had much lower phylogenetic signals than mitochondrial data; however, diagnostic substitutions for *C. t. altaicus* and *C. t. sungaris* delimitation were also revealed. Analysis of a hardly accessible report on cross experiments allowed us to conclude the partial postzygotic isolation between these subspecies. Taken together, the molecular-genetic, geographical, morphological, and reproductive isolation findings verified that *C. t. altaicus* and *C. t. sungaris* subspecies could be considered as different species.

Keywords: *Citellophilus tesquorum*; COI; flea; ITS1; ITS2; plague; population; subspecies; *Wolbachia*

Medvedev, S. G., Sedikhin, N. V., & Krasnov, B. R. (2020). Intraspecific variation of body size in fleas:

Effects of host sex and flea phenology. *Parasitology Research*, 119(10), 3211-3220.
doi:10.1007/s00436-020-06867-1

We investigated the effects of host sex and flea phenology (estimated as periods of high versus low abundance) on individual body size in four fleas of small mammals. *Amalaraeus penicilliger* and *Ctenophthalmus uncinatus* are ectoparasites of the bank vole *Myodes glareolus*, whereas *Doratopsylla dasyncnema* and *Palaeopsylla soricis* are ectoparasites of the common shrew *Sorex araneus*. We found significant effects of host sex and phenology on the body size of all flea species, although there was no general trend in the directions of these effects. Larger *A. penicilliger* were found on female hosts, whereas larger *P. soricis* were found on male hosts. In the remaining species, larger fleas were collected from male hosts during periods of high abundance (male *C. uncinatus* and female *D. dasyncnema*) and from female hosts during periods of low abundance (male *C. uncinatus*). Regarding phenology, larger fleas were recorded during periods of either high (*A. penicilliger*, *C. uncinatus*, *D. dasyncnema*) or low (*C. uncinatus*, *P. soricis*) abundance, but this depended on flea and/or host sex. We conclude that the directions of the host sex and phenology effects varied between flea species. Furthermore, the direction of the host sex effect was mediated by the effect of phenology and vice versa.

Medvedev, S. G., D. B. Verzhutsky, B. K. Kotti. 2021. DIVERSITY OF PLAGUE VECTORS: FLEAS OF THE GENUS *FRONTOPSYLLA* WAGNER ET IOFF, 1926 (SIPHONAPTERA, LEPTOPSYLLIDAE). *Entomological Review* 101(9): 1273-1286.

ПАРАЗИТОЛОГИЯ, 2021, том 55, № 6, с. 476–495. УДК 576.895.775:574.9/579.842.23, DOI: 10.31857/S003118472106003X (In Russian)

Keywords: fleas, Siphonaptera, species vectors of plague pathogen, taxonomic diversity, *Frontopsylla*

Taxonomic diversity and peculiarities of distribution and host-parasite relations of fleas of the Palearctic genus *Frontopsylla* (Leptopsyllidae, Paradoxopsyllinae) are analyzed in the present review. The role of separate representatives of this genus as pathogen vectors and reservoirs in natural plague foci of Eurasia is observed. It is demonstrated that 11 out of 36 flea species of the genus *Frontopsylla* are marked as main, secondary, or occasional vectors of this infection.

Medvedev, S. G., Verzhutsky, D. B., & Kotti, B. K. (2021). Diversity of Fleas—Plague vectors: Species of the genus *Neopsylla* Wagner, 1903 (Siphonaptera, Hystrichopsyllidae). *Entomological Review*, 101(5), 610-624. doi:10.1134/S0013873821050031

The taxonomic diversity, distribution patterns, and host-parasite relationships of fleas of the Holarctic–Indo-Malayan genus *Neopsylla* (Hystrichopsyllidae, Neopsyllinae) are analyzed. The role of some members of this genus as vectors and reservoirs in natural plague foci of Eurasia is analyzed; 14 out of 67 species and subspecies of the genus *Neopsylla* have been recorded as the main, accessory or occasional vectors of the plague pathogen.

Napoli, E., Remesar, S., Gaglio, G., Giannetto, S., Spadola, F., Díaz, P., Morrondo, P., Brianti, E. (2021) Ectoparasites of wild rabbit (*Oryctolagus cuniculus*) in Southern Italy. *Veterinary Parasitology: Regional Studies and Reports*, 24, art. no. 100555. DOI: 10.1016/j.vprsr.2021.100555

The European wild rabbit *Oryctolagus cuniculus* is a widespread mammal that may act as host for ectoparasites and reservoir for some vector-borne pathogens. Study aims were to investigate the arthropod fauna that may infect the European wild rabbit, to assess the ecological indexes and to investigate tick preferred attachment sites. The survey was conducted on 105 wild rabbits hunted by poachers in a Mediterranean woodland area in the province of Ragusa. Animals were individually examined for ectoparasite presence and the tick load assessed on eleven body anatomical sites. Ticks and fleas were found on 79 (75.2%) and 3 (2.9%) out of the 105 examined rabbits, respectively. Three tick species were identified; *Rhipicephalus pusillus* was the most abundant followed by *Rhipicephalus turanicus* and *Ixodes gibbosus*. Although ticks were found on all anatomical sites, ears, forelimbs, and abdomen were those more frequently infected. All the collected fleas were identified as *Spilopsyllus cuniculi*. Some of the ectoparasite species found on examined wild rabbits are recognized as potential vectors of pathogens of animal and human concern, such as, *Rickettsia sibirica mongolitimonae* and *Rickettsia massiliae*. The most abundant tick species found in this study is the main vector of some zoonotic pathogens; therefore, the possible role of *O. cuniculus* in their maintenance should be further investigated.

Waindok, P., Raue, K., Grilo, M.L., Siebert, U., Strube, C. (2021) Predators in northern Germany are reservoirs for parasites of One Health concern. *Parasitology Research*, 120 (12), pp. 4229-4239. DOI: 10.1007/s00436-021-07073-3

Urbanisation and invasion of wildlife into urban areas as well as human leisure activities create diverse wildlife-domestic animal-human interfaces, increasing the risk of (zoonotic) parasite spillover from sylvatic to domestic and synanthropic cycles. This study investigated the endo- and ectoparasite fauna, emphasising on parasites of One Health Concern, of the most common predators in northern Germany between November 2013 and January 2016. Eighty red foxes (*Vulpes vulpes*), 18 stone martens (*Martes foina*) and nine raccoon dogs (*Nyctereutes procyonoides*) were available for the study. Overall, 79 (73.8%) of the examined predators (n=107) harboured at least one endoparasite. The most frequently detected endoparasites in red foxes were *Toxocara canis* (43.8% positive individuals), *Capillaria* spp. (36.3%), *Alaria alata* (25.0%), *Echinococcus multilocularis* (26.3%) and *Uncinaria stenocephala* (25.0%). Furthermore, *Toxascaris leonina*, *Trichuris vulpis*, *Taenia* spp., *Mesocestoides* spp. and coccidian oocysts were observed. The endoparasite species richness in raccoon dogs was comparable to red foxes, while in stone martens, only *Capillaria* spp. were found. Muscle digestion for detection of *Trichinella* spp. and antigen testing for *Giardia* spp. did not show positive results. Ectoparasite analyses revealed infestations with ticks species of the genus *Ixodes* as well as *Dermacentor reticulatus*. Scabies mites were not present in digested skin samples, while *Demodex* spp. mites were observed by faecal flotation in one red fox. Furthermore, fleas (*Archaeopsylla erinacei* and *Chaetopsylla globiceps*) were observed in the fur of red foxes, while lice were not present in any predator species. However, infestation frequency with ectoparasites was with 19.2% generally low in available predator skins (n=99). Overall, the present study showed that predators in northern Germany serve as reservoirs for parasites of One Health concern, with four of the five most frequent endoparasites being zoonotic, highlighting the need of parasite surveillance in wildlife predators in order to implement measures avoiding spillovers to domestic animals and humans.

Physiology

Buchberger, G., Kogler, A., Weth, A., Baumgartner, R., Comanns, P., Bauer, S., Baumgartner, W.

(2018) Erratum to: “Fluidic Diode” for Passive Unidirectional Liquid Transport Bioinspired by the Spermathecae of Fleas (Journal of Bionic Engineering, 15, 1, (42-56), 10.1007/s42235-017-0003-7)
 (2021) Journal of Bionic Engineering, 18 (3), p. 733. DOI: 10.1007/s42235-018-0031-y

Deng, Z., Zeng, Q., Tang, J., Zhang, B., Chai, J., Andersen, J.F., Chen, X., Xu, X. (2021) Anti-inflammatory effects of FS48, the first potassium channel inhibitor from the salivary glands of the flea *Xenopsylla cheopis*. Journal of Biological Chemistry, 296, art. no. 100670. DOI: 10.1016/j.jbc.2021.100670

The voltage-gated potassium (Kv) 1.3 channel plays a crucial role in the immune responsiveness of T-lymphocytes and macrophages, presenting a potential target for treatment of immune- and inflammation related-diseases. FS48, a protein from the rodent flea *Xenopsylla cheopis*, shares the three disulfide bond feature of scorpion toxins. However, its three [-] dimensional structure and biological function are still unclear. In the present study, the structure of FS48 was evaluated by circular dichroism and homology modeling. We also described its in vitro ion channel activity using patch clamp recording and investigated its anti-inflammatory activity in LPS-induced Raw 264.7 macrophage cells and carrageenan [-] induced paw edema in mice. FS48 was found to adopt a common $\alpha\beta$ structure and contain an atypical dyad motif. It dose-dependently exhibited the Kv1.3 channel in Raw 264.7 and HEK 293T cells, and its ability to block the channel pore was demonstrated by the kinetics of activation and competition binding with tetraethylammonium. FS48 also downregulated the secretion of proinflammatory molecules NO, IL-1 β , TNF- α , and IL-6 by Raw 264.7 cells in a manner dependent on Kv1.3 channel blockage and the subsequent inactivation of the MAPK/NF- κ B pathways. Finally, we observed that FS48 inhibited the paw edema formation, tissue myeloperoxidase activity, and inflammatory cell infiltrations in carrageenan [-] treated mice. We therefore conclude that FS48 identified from the flea saliva is a novel potassium channel inhibitor displaying anti-inflammatory activity. This discovery will promote understanding of the bloodsucking mechanism of the flea and provide a new template molecule for the design of Kv1.3 channel blockers.

Liu, T., Ou, J.-Y., MacDonald, K.F., Zheludev, N.I. (2021) Detection of sub-atomic movement in nanostructures. Nanoscale Advances, 3 (8), pp. 2213-2216. DOI: 10.1039/d0na01068e

Nanoscale objects move fast and oscillate billions of times per second. Such movements occur naturally in the form of thermal (Brownian) motion while stimulated movements underpin the functionality of nano-mechanical sensors and active nano-(electro/opto) mechanical devices. Here we introduce a methodology for detecting such movements, based on the spectral analysis of secondary electron emission from moving nanostructures, that is sensitive to displacements of sub-atomic amplitude. We demonstrate the detection of nanowire Brownian oscillations of ~ 10 pm amplitude and hyperspectral mapping of stimulated oscillations of setae on the body of a common flea. The technique opens a range of opportunities for the study of dynamic processes in materials science, nanotechnology and biology.

Lu, S., Tirloni, L., Oliveira, M.B., Bosio, C.F., Nardone, G.A., Zhang, Y., Hinnebusch, B.J., Ribeiro, J.M., Andersen, J.F. (2021) Identification of a substrate-like cleavage-resistant thrombin inhibitor from the saliva of the flea *Xenopsylla cheopis*. Journal of Biological Chemistry, 297 (5), art. no. 101322, . Cited 1 time. DOI: 10.1016/j.jbc.2021.101322

The salivary glands of the flea *Xenopsylla cheopis*, a vector of the plague bacterium, *Yersinia pestis*, express proteins and peptides thought to target the hemostatic and inflammatory systems of its mammalian hosts. Past transcriptomic analyses of salivary gland tissue revealed the presence of two similar peptides (XC-42 and XC-43) having no extensive similarities to any other deposited sequences. Here we show that these peptides specifically inhibit coagulation of plasma and the amidolytic activity of α -thrombin. XC-43, the smaller of the two peptides, is a fast, tight-binding inhibitor of thrombin with a dissociation constant of less than 10 pM. XC-42 exhibits similar selectivity as well as kinetic and binding properties. The crystal structure of XC-43 in complex with thrombin shows that despite its substrate-like binding mode, XC-43 is not detectably cleaved by thrombin and that it interacts with the thrombin surface from the enzyme catalytic site through the fibrinogen-binding exosite I. The low rate of hydrolysis was verified in solution experiments with XC-43, which show the substrate to be largely intact after 2 h of incubation with thrombin at 37 °C. The low rate of XC-43 cleavage by thrombin may be attributable to specific changes in the catalytic triad observable in the crystal structure of the complex or to extensive interactions in the prime sites that may stabilize the binding of cleavage products. Based on the increased arterial occlusion time, tail bleeding time, and blood coagulation parameters in rat models of thrombosis XC-43 could be valuable as an anticoagulant.

Evolution

Golas, B.D., Goodell, B., Webb, C.T. (2021) Host adaptation to novel pathogen introduction: Predicting conditions that promote evolutionary rescue. *Ecology Letters*, 24 (10), pp. 2238-2255. DOI: 10.1111/ele.13845

Novel pathogen introduction can have drastic consequences for naive host populations, and outcomes can be difficult to predict. Evolutionary rescue (ER) provides a foundation for understanding whether hosts are driven to extinction or survive via adaptation. Currently, patterns of host population dynamics alongside evidence of adaptation are used to infer ER. However, the gap between established ER theory and complexity inherent in natural systems makes interpreting empirical patterns difficult because they can be confounded with ecological drivers of survival under current theory. To bridge this gap, we expand ER theory to include biological selective agents, such as pathogens. We find birth processes to be more important than previously theorised in determining ER potential. We employ a novel framework evaluating ER potential within natural systems and gain ability to identify system characteristics that make ER possible. Identifying these characteristics allows a shift from retrospective observation to a predictive mindset, and our findings suggest that ER occurrence may be more limited than previously thought. We use the plague system of *Yersinia pestis* infecting *Cynomys ludovicianus* (black-tailed prairie dogs) and *Spermophilus beecheyi* (California ground squirrels) as a case study.

Suntsov, V.V. (2021) Genomogenesis of the Plague Bacteria *Yersinia pestis* as a Process of Mosaic Evolution. *Russian Journal of Genetics*, 57 (2), pp. 139-151. DOI: 10.1134/S1022795421020113

The guiding methodological principle of the modern molecular genetic (MG) approach to the problem of speciation of the plague bacteria *Yersinia pestis* is the saltationist paradigm of horizontal gene transfer. The main speciation events include the insertion of two virulence plasmids specific to *Y. pestis*, pFra and pPst, into the genome of ancestral psychrophilic saprozoonotic pseudotuberculosis bacteria *Y. pseudotuberculosis* O:1b, as well as inactivation/deletion of genes that have lost their functions. An alternative ecological approach suggests the “Darwinian” mechanism of the plague bacterium genome formation in the temperature-continuous (5–37°C) environment, i.e., the Tarbagan

marmot–marmot flea (*Marmota sibirica*–*Oropsylla silantiewi*) parasitic system, which demonstrates intermediate properties between the habitats of progenitor and derivative species. In accordance with the ecological scenario of the plague origin, the evolutionary fate of 18 well-studied plague bacterium genes and gene structures was interpreted. It was demonstrated that gradual transformation of the genome of the pseudotuberculosis bacterium into the genome of plague causative agent proceeded according to the principle of mosaic evolution and corresponded to three types of adaptation, i.e., inadaptation, pre-adaptation and neo-adaptation. The prospects for further development of the theory of plague origin are thought to be in the synthesis of MG and ecological approaches.

Tihelka, E., Cai, C., Giacomelli, M., Lozano-Fernandez, J., Rota-Stabelli, O., Huang, D., Engel, M.S., Donoghue, P.C.J., Pisani, D. (2021) The evolution of insect biodiversity. *Current Biology*, 31 (19), pp. R1299-R1311. DOI: 10.1016/j.cub.2021.08.057

Insects comprise over half of all described animal species. Together with the Protura (coneheads), Collembola (springtails) and Diplura (two-pronged bristletails), insects form the Hexapoda, a terrestrial arthropod lineage characterised by possessing six legs. Exponential growth of genome-scale data for the hexapods has substantially altered our understanding of the origin and evolution of insect biodiversity. Phylogenomics has provided a new framework for reconstructing insect evolutionary history, resolving their position among the arthropods and some long-standing internal controversies such as the placement of the termites, twisted-winged insects, lice and fleas. However, despite the greatly increased size of phylogenomic datasets, contentious relationships among key insect clades remain unresolved. Further advances in insect phylogeny cannot rely on increased depth and breadth of genome and taxon sequencing. Improved modelling of the substitution process is fundamental to countering tree-reconstruction artefacts, while gene content, modelling of duplications and deletions, and comparative morphology all provide complementary lines of evidence to test hypotheses emerging from the analysis of sequence data. Finally, the integration of molecular and morphological data is key to the incorporation of fossil species within insect phylogeny. The emerging integrated framework of insect evolution will help explain the origins of insect megadiversity in terms of the evolution of their body plan, species diversity and ecology. Future studies of insect phylogeny should build upon an experimental, hypothesis-driven approach where the robustness of hypotheses generated is tested against increasingly realistic evolutionary models as well as complementary sources of phylogenetic evidence.

Tschirren, B., Romero-Haro, A.Á., Zahn, S., Criscuolo, F. (2021) Sex-specific effects of experimental ectoparasite infestation on telomere length in great tit nestlings. *Journal of Evolutionary Biology*, 34 (3), pp. 584-589. DOI: 10.1111/jeb.13744

Telomere length is a biomarker of biological ageing and lifespan in various vertebrate taxa. Evidence is accumulating that telomeres shorten more rapidly when an individual is exposed to environmental stressors. Parasites are potent selective agents that can cause physiological stress directly or indirectly through the activation of the host's immune system. Yet to date, empirical evidence for a role of parasites in telomere dynamics in natural populations is limited. Here, we show experimentally that exposure to ectoparasitic hen fleas (*Ceratophyllus gallinae*) during growth results in shorter telomeres in female, but not male, great tit (*Parus major*) nestlings. Females had longer telomeres than males when growing up in experimentally deparasitized nests but, likely because of the sex-specific effects of ectoparasitism on telomere length, this sexual dimorphism was absent in birds growing up in experimentally infested nests. Our results provide the first experimental evidence for a

role of ectoparasitism in telomere dynamics in a natural vertebrate population, and suggest that the costs of infection manifest in sex-specific ways.

Ecology

Baláz, I., Ševčík, M., Tulis, F., Zigová, M., Dudich, A. (2021) Diversity, distribution and changes in communities of fleas on small mammals along the elevational gradient from the Pannonian Plain to the Carpathian Mountains. *Parasitology*, 148 (1), pp. 63-73. DOI: 10.1017/S0031182020002024

The causal chain of parasite-host-environment interactions, the so-called 'dual parasite environment', makes studying parasites more complicated than other wild organisms. A sample of 65 282 fleas taken from 336 different locations were analysed for changes in the distribution, diversity and compensation of flea communities found on small mammals along an elevational diversity gradient ranging from the Pannonian Plain to the base of the Carpathian summits. The fleas were divided into four groups, which were derived from changes in abundance and occurrence determined from cluster analysis. They are (1) flea species whose range seems unrelated to any change in elevation; (2) species that avoid high altitudes; (3) a group that can be subdivided into two types: (i) host-specific fleas and (ii) mountains species and (4) species opting for high altitudes on the gradient or preferring lower to middle elevations below 1000 m. Our study showed a unimodal pattern of flea diversity along the elevational gradient. It indicated that seasonality significantly conditions changes in biodiversity and patterns of spatial change along the elevational gradient, with specific flea species influenced by their host, while the impact of environmental conditions is more pronounced in opportunistic flea species.

Castano-Vázquez, F., Schumm, Y.R., Bentele, A., Quillfeldt, P., Merino, S. (2021) Experimental manipulation of cavity temperature produces differential effects on parasite abundances in blue tit nests at two different latitudes. *International Journal for Parasitology: Parasites and Wildlife*, 14, pp. 287-297. DOI: 10.1016/j.ijppaw.2021.03.010

Although different predictive models forecast that climate change will alter the distribution and incidence of parasitic diseases, few studies have investigated how microclimatic changes may affect host-parasite relationships. In this study, we experimentally increased the temperature inside nest boxes of the blue tit *Cyanistes caeruleus* during the nestling period at two different latitudes (central Spain and central Germany) to determine its effect on parasite abundance. The two localities have contrasting climate conditions: the southern one in Spain is warmer and drier than the northern one in Germany. Consistent with this, we observed that the parasitic fauna in nests at the two localities differs. The flea species *Ceratophyllus gallinae* was more abundant in the northern locality, while the blowfly species *Protocalliphora azurea* and biting midge species of the genus *Culicoides* were more abundant in the southern one, as were blood parasites. Moreover, dermanyssid mites and blackflies (Simuliidae) were observed only in the southern locality. The temperature inside nest boxes was increased using heat mats placed underneath the nest material during the nestling period (day 3 to day13 post-hatching). Compared with control nests, the average temperature in heated nests increased by 2.24 °C and 1.35 °C at night in Spain and Germany, respectively. Consequently, the average relative humidity in heated versus control nests decreased 4.93 and 0.82 units in Spain and Germany, respectively. The abundance of blowfly pupae in the heated nests was significantly lower than that of control nests at both localities. The abundance of larval fleas was also lower in the heated nests, but only at the Spanish locality. Infection by the blood parasites *Haemoproteus/Plasmodium* was higher in males attending the heated nests in Germany, and the control nests in Spain. Moreover, both male body mass and nestling wing

length were negatively related to the abundance of larval fleas. In conclusion, our results indicate that increased temperature at the nestling stage may affect the fitness of blue tits by altering parasite prevalence rates.

Di Giovanni, F., Wilke, A.B.B., Beier, J.C., Pombi, M., Mendoza-Roldan, J.A., Desneux, N., Canale, A., Lucchi, A., Dantas-Torres, F., Otranto, D., Benelli, G. (2021) Parasitic strategies of arthropods of medical and veterinary importance. *Entomologia Generalis*, 41 (5), pp. 511-522. DOI: 10.1127/entomologia/2021/1155

Arthropoda is the animal phylum that includes the largest number of animal species on Earth. Several arthropods are of medical and veterinary significance in terrestrial ecosystems, some of them playing a crucial role in the transmission of pathogens, which may infect and eventually cause diseases in a wide range of vertebrates, including humans, pets, live-stock, and wildlife. Nevertheless, a clear categorization of the parasitic strategies carried out by arthropods of medical and veterinary interest is still lacking. Herein, according to an earlier discussion within the scientific community and based on the parasitic strategy implemented, we suggest that terrestrial parasitic arthropods of medical and veterinary interest should be classified into three categories:

- (i) trophically transmitted parasites (pentastomids),
- (ii) directly transmitted parasites (sucking lice, chewing lice, itch mites, and skin mites) and
- (iii) micropredators, with a further division within the category micropredators to distinguish between
 1. short-term (blood-sucking or secretophagous flies, bed bugs, kissing bugs, fleas, and ticks) and
 2. long-term (chigoe fleas and myiasis-causing larvae) micropredators.

A thorough understanding and consensual categorization of the parasitic strategies adopted by arthropods of medical and veterinary significance may pave the way for future multidisciplinary basic to applied research.

Finlayson, G., Taggart, P., Cooke, B. (2021) Recovering Australia's arid-zone ecosystems: learning from continental-scale rabbit control experiments. *Restoration Ecology*. DOI: 10.1111/rec.13552

Introduced rabbits are a continuing threat to native Australian flora and fauna. Three interventions using biological control agents, myxomatosis, European rabbit fleas, and rabbit hemorrhagic disease, have reduced rabbit abundance and kept numbers low over the last 70 years. We considered the benefits of biological control for native fauna to put the role of rabbits in influencing vegetation cover, food supply, and predation into better perspective. Numerous examples exist demonstrating increases in native vegetation and the expansion and recovery of native animal populations at landscape scales following intense rabbit suppression. Ongoing research on methods for supplementing the impact of biological control agents and managing introduced predators are needed to restore Australia's arid-zone ecosystems. However, many biologists and rangeland managers first need to reevaluate the misconception that removing rabbits also introduces other serious and insurmountable problems such as prey-switching by introduced cats and foxes.

- Fantozzi, M. C., J. Sanchez; M. Lareschi; P.M. Beldomenico. 2021. Effects of host factors on the dynamics of fleas (Siphonaptera) in Sigmodontinae rodents (Cricetidae) from El Espinal Ecoregion, Argentina. *Acta Tropica*. <https://doi.org/10.1016/j.actatropica.2021.106177>
- Gozzi A.C., Marcela Lareschi, G. Navone, M.L. Guichón. 2020. The Enemy Release Hypothesis and *Callosciurus erythraeus* in Argentina: combining community and biogeographical parasitological studies. *Biological Invasions* 22 (12). DOI 10.1007/s10530-020-02339-w
- Krasnov, B. M. Stanko, M. Lareschi, I. S. Khokhlova. 2020. Species co-occurrences in ectoparasite infracommunities: accounting for confounding factors associated with space, time, and host community composition. *Ecological Entomology* 45: 1158-1171. DOI:10.1111/een.12900
- Lareschi M., S. González, M. Urdapilleta, M. Melis, E. Savchenko, M. Espinoza-Carniglia & A. González. 2021. ¿Qué sabemos de las pulgas? Experiencias de extensión universitaria con niños y niñas en ámbitos escolares III Jornadas Argentinas de Etnobiología y Sociedad. 10, 11 y 12 de noviembre de 2021, La Plata - Modalidad virtual. *Revista del Museo de La Plata*, Vol 6, Suplemento Resúmenes: 1R-264R: 239
- Melis M., M. Espinoza-Carniglia, E. Savchenko, S. Nava, M. Lareschi. 2020. Molecular detection and identification of *Rickettsia felis* in a *Polygenis* flea (Siphonaptera, Rhopalopsyllidae, Rhopalopsyllinae) associated with cricetid rodents from a rural area in central Argentina. *Veterinary Parasitology: Regional Studies and Reports* 21 (2020): 100445 <https://doi.org/10.1016/j.vprsr.2020.100445>
- Melis, M. E., Sebastian, P.S., Balcazar, D.E., Lareschi, M. & S. Nava. 2020. Molecular detection of *Rickettsia bellii* in *Ixodes loricatus* (Acari: Ixodidae) ticks associated with rodents from Buenos Aires Province, Argentina. *Revista de la Sociedad Entomológica Argentina* 79 (4): 51-55 DOI <https://doi.org/10.25085/rsea.790409>
- Melis, M.E., Savchenko, E., Urdapilleta, M., Espinoza-Carniglia, M., Giambelluca, L., Balcazar, D., Galliari, C., Nava, S., Lareschi, M. 2021. La comunidad componente de ectoparásitos en roedores sigmodontinos (Cricetidae) del Gran La Plata, Argentina: Prospección de *Rickettsia* spp. en los artrópodos. *Jornadas Argentinas de Mastozoología Virtuales*, 16 al 19 de Noviembre 2021. Pág. 155
- Moreno Salas L., M. Espinoza-Carniglia, N. Lizama-Schmeisser, L. G. Torres Fuentes, C. Silva de La Fuente, M. Lareschi and D. González-Acuña. 2020. Molecular detection of *Rickettsia* in fleas from micromammals in Chile. *Parasite & Vectores*, 13: 523. DOI: 10.21203/rs.3.rs-35524/v1
- Urdapilleta M., Cicuttin GL, De Salvo MN, Pech May A, Salomon OD & Lareschi, M. 2020. Molecular detection and identification of *Bartonella* in the cat flea *Ctenocephalides felis felis* collected from cats and dogs in a border area in northeastern Argentina. *Veterinary Parasitology: Regional Studies and Reports* 19: 100361 <https://doi.org/10.1016/j.vprsr.2019.100361>.
- Urdapilleta M., Galliari, C.A., Navarro Febre, T, & M.Lareschi. 2022. Effect of host and environment related factors on the distribution of the ectoparasites of the montane grass mouse *Akodon montensis* (Cricetidae, Sigmodontinae) in the Atlantic Forest ecoregion in northeastern Argentina. *Revista Mexicana de Biodiversidad*. In Press.

Urdapilleta M., Pech-May A.; Lamattina D.; Burgos F.E.; Balcazar E.D.; Ferrari W.A.; Lareschi M. & O.D. Salomón. 2021. Ecology of fleas and their hosts in the triffinio of northeast Argentina: First detection of *Rickettsia asembonensis* in *Ctenocephalides felis felis* in Argentina. Medical and Veterinary Entomology. <https://onlinelibrary.wiley.com/doi/10.1111/mve.12546>

Urdapilleta, M., Burgos, F.E., Lamattina, D., Giuliani, M., Cortés, M., Meichtry, B., Balcázar, D., De Salvo, M.N., Cicuttin, L.G., Salomón, D.O., Lareschi, M. 2021. Vectores y reservorios de bacterias patógenas: *Bartonella* en mamíferos silvestres y sus pulgas asociadas en la Selva Paranaense Argentina. Jornadas Argentinas de Mastozoología Virtuales, 16 al 19 de Noviembre 2021. Pág. 157.

Urdapilleta, M., Lamattina, D., Pech-May, A., Burgos E.F., Balcazar E. D., Ferrari, A.O. W., Giuliani M., Cortés M.M., Meichtry B., Salomón, O.D1 & Lareschi M. 2021. Primera detección de *Rickettsia asembonensis* en *Ctenocephalides felis felis* en Argentina. Un estudio epidemiológico en pulgas de animales de compañía y sinantrópicos en el triffinio del noreste argentino. I Congreso Virtual de Microbiología Veterinaria, FCV, UNLP. Del 4 al 6 de agosto de 2021.

Pathology and Control

Barker, H. (2021) Laying the corpses to rest: Grain, embargoes, and *Yersinia pestis* in the black sea, 1346–48. Speculum, 96 (1), pp. 98-126. DOI: 10.1086/711596

When, how, and why did the Black Death reach Europe? Historians have relied on Gabriele de' Mussi's account of Tatars catapulting plague-infested bodies into the besieged city of Caffa on the Crimean Peninsula. Yet Mussi spent the 1340s in Piacenza; he had no direct knowledge of events in Caffa. Sources by people present in the Black Sea during the Second Pandemic, including Genoese colonial administrators, Venetian diplomats, Byzantine chroniclers, and Mamluk merchants, offer a different perspective. They show that the Venetian community at Tana played an important role in plague transmission; that it took over a year (from spring 1346 to autumn 1347) for plague to cross the Black Sea to Constantinople; that people crossed the Black Sea in 1346 but commodities did not because of a series of trade embargoes; that grain was one of the most important Black Sea commodities in both volume and strategic value; and therefore that the embargoes of 1346 delayed plague transmission by temporarily halting the movement of grain with its accompanying rats, fleas, and bacteria. When Venice, Genoa, and the Golden Horde made peace and lifted their embargoes in 1347, both the grain trade and the spread of plague resumed.

Di Tommaso, M., Luciani, A., Crisi, P.E., Beschi, M., Rosi, P., Rocconi, F., Miglio, A. (2021) Detection of serum allergen-specific IgE in atopic dogs tested in northern Italy: Preliminary study. Animals, 11 (2), art. no. 358, pp. 1-12. DOI: 10.3390/ani11020358

Canine atopic dermatitis (CAD) is a pruritic allergic skin disease associated with IgE-mediated hypersensitivity. IgE is detected using Serum Allergen-Specific IgE test (SAT) in order to identify allergens. The present study aims to identify the environmental allergens in atopic dogs living in Northern Italy using SAT. The screening SAT (sSAT), using a monoclonal antibody cocktail-based ELISA to identify indoor and outdoor allergens, was performed. In all positive samples, an anti-IgE

monoclonal antibody ELISA test was performed to extend panel of allergens. Out of 117 selected dogs, 69 were included in the study; 71% were positive and 29% were negative to sSAT. Among the 49 positive sSAT, 53% were positive for both indoor and outdoor, 38.8% only for indoor, and 8.2% only for outdoor allergens. This is the first study on the frequency of allergens involved in CAD in Italy using SAT. IgE hypersensitivity in atopic dogs of Northern Italy is usually associated with indoor allergens, primarily house dust mites. Among the outdoor allergens, an important role was played by *Rumex acetosa* [Editor's note: common sorrel]. Polysensitization also commonly occurs. Therefore, since the numerous factors affect the IgE positivity in CAD, specific panels for geographical areas should be considered and re-evaluated at time intervals.

Leite, L.L., Bakos, R.M. (2021) Visual Dermatology: Flea Movements Demonstrated by Dermoscopy. *Journal of Cutaneous Medicine and Surgery*, 25 (2), p. 209. DOI: 10.1177/1203475420929922

Perkins, R., Whitehead, M., Civil, W., Goulson, D. (2021) Potential role of veterinary flea products in widespread pesticide contamination of English rivers. *Science of the Total Environment*, 755, art. no. 143560. DOI: 10.1016/j.scitotenv.2020.143560

Little is known about the environmental fate or impact of pesticides used to control companion animal parasites. Using data from the Environment Agency, we examined the occurrence of fipronil, fipronil metabolites and imidacloprid in 20 English rivers from 2016 to 2018, as indicators of the potential contamination of waterways from their use as ectoparasiticides on pets. Water samples were collected by the Environment Agency as part of their chemical surveillance programme and analysed using Liquid Chromatography Mass Spectrometry / Quadrupole-Time-of-Flight Mass spectrometry (LC/Q-TOF-MS) methods. A total of 3861 chemical analyses were examined, and the significance and potential sources of this contamination were assessed. Fipronil, fipronil sulfone, fipronil sulfide (collectively known as fiproles) and imidacloprid were detected in 98.6%, 96.5%, 68.7% and 65.9% of samples, respectively. Across the river sites sampled, the mean concentrations of fipronil (17 ng/l, range <0.3–980 ng/l), and fipronil sulfone (6.5 ng/l, range <0.2–39 ng/l) were 5.3 and 38.1 times their chronic toxicity limits of 3.2 and 0.17 ng/l, respectively. Imidacloprid had a mean concentration of 31.7 ng/l (range <1–360 ng/l), which was below its chronic toxicity limit of 35 ng/l, however seven out of 20 sites exceeded that limit. Chronic risk quotients indicate a high environmental risk to aquatic ecosystems from fiproles, and a moderate risk from imidacloprid. Sites immediately downstream of wastewater treatment works had the highest levels of fipronil and imidacloprid, supporting the hypothesis that potentially significant quantities of pesticides from veterinary flea products may be entering waterways via household drains. These findings suggest the need for a reevaluation of the environmental risks associated with the use of companion animal parasiticide products, and the risk assessments that these products undergo prior to regulatory approval.

López-Pérez, A.M., Chaves, A., Sánchez-Montes, S., Foley, P., Uhart, M., Barrón-Rodríguez, J.,

Becker, I., Suzán, G., Foley, J. (2021) Diversity of rickettsiae in domestic, synanthropic, and sylvatic mammals and their ectoparasites in a spotted fever-epidemic region at the western US-Mexico border. *Transboundary and Emerging Diseases*. DOI: 10.1111/tbed.14027

Over one hundred cases of human rickettsiosis, many fatal, are reported annually across the US-Mexico transboundary region, representing a likely undercount. Although cases are often attributed to *Rickettsia rickettsii*, the agent of Rocky Mountain spotted fever, multiple other *Rickettsia* pathogens are present in North America. We conducted multiple-host surveillance of domestic, synanthropic, and sylvatic mammals and their ectoparasites to investigate the ecology of *Rickettsia* species in this region. A total of 499 mammals, including 83 dogs, 23 wild carnivores, five lagomorphs, and 388 rodents were sampled, and 413 fleas and 447 ticks belonging to 15 and 4 species, respectively, were collected during 2017 and 2018. We detected *Rickettsia* spp. DNA in one blood sample of coyote (*Canis latrans*), 11 ear tissues of rodents (10.6%), and 79 ectoparasites (9.5%). Of the 64 *Rickettsia*-positive fleas, 54 were *Echidnophaga gallinacea* and 10 were *Pulex simulans*, while of the 15 ticks, 11 were *Rhipicephalus sanguineus* s.l. and four *Ixodes pacificus*. The DNA sequence alignment of *gltA* and *ompB* regions revealed one and ten genetic variants of *Rickettsia* spp., respectively. These variants were clustered in clades of zoonotic species (*R. felis*, *R. massiliae*, *R. parkeri*, *R. rickettsii*, and *R. typhi*) and organisms of unknown pathogenic significance (*R. asembonensis* and *Candidatus Rickettsia tarasevichiae*). The finding of a coyote infected with *R. rickettsii* and the multiple zoonotic SFG rickettsial agents in the study area suggest that: 1) wild canids could serve as an amplifying host for RMSF, an alternate host for *Rh. sanguineus* s.l. ticks, and a means to spread infection and ticks over large areas; and 2) at least some of the human rickettsiosis cases attributed to *R. rickettsii* could be caused by other *Rickettsia* species. This study strongly supports the importance of multiple-host and vector eco-epidemiological studies and the One Health approach to better understand disease in a RMSF-epidemic region.

Microbial Symbiotes

Benelli, G., Wassermann, M., Brattig, N.W. (2021) Insects dispersing taeniid eggs: Who and how? (2021) *Veterinary Parasitology*, 295, art. no. 109450. DOI: 10.1016/j.vetpar.2021.109450

Taeniosis/cysticercosis and echinococcosis are neglected zoonotic helminth infections with high disease burden caused by tapeworms which circulate between definitive and intermediate host reflecting a predator-prey interaction. Taeniid eggs can remain vital for months, allowing arthropods to mechanically transport them to intermediate hosts. However, the multiple routes that arthropods provide as carriers of taeniid eggs are still often unregarded or not considered. This review focuses on the prevalence and importance of arthropods as carriers and spreaders of taeniid eggs in the epidemiology of taeniosis/cysticercosis and echinococcosis. Current scientific knowledge showed a relevant role of houseflies (Muscidae), blowflies (Calliphoridae), dung beetles (Scarabaeoidea), darkling beetles (Tenebrionidae), ground beetles (Carabidae) and skin beetles (Dermestidae) in the spread of taeniid eggs in the environment, which may favor the infection of new hosts through the direct ingestion of an insect or of contaminated food and water. At last, key research challenges are highlighted, illustrating that further knowledge on the topic is needed to develop and improve guidelines and actions to prevent taeniid infections worldwide.

[Editor's note: some fleas carry tapeworms (e.g. the cat and dog flea tapeworm *Dipylidium caninum*, family Dipylidiidae), but it is unclear to me if fleas ever carry Taeniidae.]

Flatau, R., Segoli, M., Hawlena, H. (2021) *Wolbachia* Endosymbionts of Fleas Occur in All Females but Rarely in Males and Do Not Show Evidence of Obligatory Relationships, Fitness Effects, or Sex-Distorting Manipulations. *Frontiers in Microbiology*, 12, art. no. 649248, DOI: 10.3389/fmicb.2021.649248

The widespread temporal and spatial persistence of endosymbionts in arthropod host populations, despite potential conflicts with their hosts and fluctuating environmental conditions, is puzzling. Here, we disentangled three main mechanisms that are commonly proposed to explain such persistence, namely, obligatory relationships, in which the host is fully dependent on its endosymbiont, fitness advantages conferred by the endosymbiont, and reproductive manipulations imposed by the endosymbiont. Our model system reflects an extreme case, in which the *Wolbachia* endosymbiont persists in all female flea hosts but rarely in male ones. We cured fleas of both sexes of *Wolbachia* but found no indications for either lower reproduction, offspring survival, or a change in the offspring sex ratio, compared to *Wolbachia*-infected fleas. These results do not support any of the suggested mechanisms. We highlight future directions to advance our understanding of endosymbiont persistence in fleas, as well as in other model systems, with extreme sex-differences in endosymbiont persistence. Insights from such studies are predicted to shed light on the evolution and ecology of arthropod-endosymbiont interactions in nature.

Book review

Richard C. Wilkerson, Yvonne-Marie Linton, and Daniel Strickman. 2021. *Mosquitoes of the World*, Volumes 1 and 2. Johns Hopkins University Press, Baltimore, Maryland, USA.

"The definitive reference on the biology, evolution, ecology, and diversity of all known species of the world's mosquitoes. Critical for entomologists, public health professionals, and epidemiologists across the world.

Biting multiple times on two, three, or more different hosts, it is no surprise that some species of mosquitoes have co-evolved with pathogens. For humans and other animals, the result has been some of the most challenging diseases known. It has been said that *Anopheles gambiae*, as the primary transmitter of malaria parasites to humans, is the most dangerous animal in the world. Certainly, malaria has killed more people than all the wars that ever took place. Even now, despite drugs and mosquito control, malaria claims the lives of 405,000 per year. The vast majority of mosquito species are not involved in pathogen transmission to humans, but those that are make a huge impact on global health.

In this two-volume set, three of the world's leading experts on mosquito disease, ecology, and systematics offer readers unique insights into the fascinating world of mosquitoes while illustrating

their diagnostic morphological features in detail. Comprehensively addressing the natural diversity of mosquitoes, the book explains their life histories, bionomic traits, and the physiological and physical adaptations they evolved in response to ever-changing environmental conditions. Mosquitoes are one of the best-known groups of insects, making this book a great starting place for anyone who would like to understand entomology by knowing the details about a representative family.

Volume One contains a review of the biology and diversity of mosquitoes. Biology is treated in the following chapters: Evolution, Nomenclature, Distribution, Development, Dormancy, Mosquito Movement, Feeding and Nutrition, Excretion, Copulation and Insemination, Egg Development and Oviposition. The chapters on biology are followed by a well-illustrated summary of the characteristics of all 41 genera and of representative species of mosquitoes. This treatment of the morphological diversity of mosquitoes is accompanied by a glossary of all morphological terms used.

Volume Two features a long-awaited comprehensive mosquito taxonomic catalog detailing the current taxonomic and systematic status of all 3,698 valid species and subspecies, 41 genera, and 187 subgenera, a list of all taxa for definitive use of nomenclature, complete lists of species synonyms, distributions, key taxonomic works, and newly defined informal names, origins of scientific names. Readers will discover that some mosquitoes undertake courtship rituals, while others guard their eggs, feed solely on earthworms, or can survive as immatures under ice sheets or in salt-encrusted pools. Hundreds of drawings and high-resolution, close-up images illustrate the text. The most complete reference work on mosquitoes ever produced, *Mosquitoes of the World* is an unmatched resource for entomologists, public health professionals, epidemiologists, and reference libraries."

- Review from Johns Hopkins University Press, press.jhu.edu/books/title/10664/mosquitoes-world

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Fleas in Art and History

"Then came Spanish 'flu among the labour force, smallpox among the herdsmen, and Delamare's cook died of bubonic plague. 'My house is overrun with plague-carrying rats,' Llewelyn wrote. 'I am now trying to catch them and in one day have caught thirty.' The fleas that transmitted the infection were only supposed to be able to jump two inches, but 'not a few have had spring enough to reach my trousers, where to my great content I have caught and killed no less than three of these naughty insects'."

pg.101, by Elspeth Josecelin Grant Huxley (1907-) 1985. Out in the midday sun: my Kenya.
Viking Publ., New York.

Belhomme's Beast
by Guy de Maupassant
(b. 5 August 1850, Tourville-sur-Arques, France - 6 July 1893, Passy, Paris)

The coach for Havre was ready to leave Criquetot, and all the passengers were waiting for their names to be called out, in the courtyard of the Commercial Hotel kept by Monsieur Malandain, Jr.

It was a yellow wagon, mounted on wheels which had once been yellow, but were now almost gray through the accumulation of mud. The front wheels were very small, the back ones, high and fragile, carried the large body of the vehicle, which was swollen like the belly of an animal. Three white horses, with enormous heads and great round knees, were the first things one noticed. They were harnessed ready to draw this coach, which had something of the appearance of a monster in its massive structure. The horses seemed already asleep in front of the strange vehicle.

The driver, Cesaire Horlaville, a little man with a big paunch, supple nevertheless, through his constant habit of climbing over the wheels to the top of the wagon, his face all aglow from exposure to the brisk air of the plains, to rain and storms, and also from the use of brandy, his eyes twitching from the effect of constant contact with wind and hail, appeared in the doorway of the hotel, wiping his mouth on the back of his hand. Large round baskets, full of frightened poultry, were standing in front of the peasant women. Cesaire Horlaville took them one after the other and packed them on the top of his coach; then more gently, he loaded on those containing eggs; finally he tossed up from below several little bags of grain, small packages wrapped in handkerchiefs, pieces of cloth, or paper. Then he opened the back door, and drawing a list from his pocket he called:

"Monsieur le cure de Gorgeville."

The priest advanced. He was a large, powerful, robust man with a red face and a genial expression. He hitched up his cassock to lift his foot, just as the women hold up their skirts, and climbed into the coach.

"The schoolmaster of Rollebose-les-Grinets."

The man hastened forward, tall, timid, wearing a long frock coat that fell to his knees, and he in turn disappeared through the open door.

"Maitre Poirot, two seats."

Poirot approached, a tall, round-shouldered man, bent by the plow, emaciated through abstinence, bony, with a skin dried by a sparing use of water. His wife followed him, small and thin, like a tired animal, carrying a large green umbrella in her hands.

"Maitre Rabot, two seats."

Rabot hesitated, being of an undecided nature. He asked:

"You mean me?"

The driver was going to answer with a jest, when Rabot dived head first towards the door, pushed forward by a vigorous shove from his wife, a tall, square woman with a large, round stomach like a barrel, and hands as large as hams.

Rabot slipped into the wagon like a rat entering a hole.

"Maitre Caniveau."

A large peasant, heavier than an ox, made the springs bend, and was in turn engulfed in the interior of the yellow chest.

"Maitre Belhomme."

Belhomme, tall and thin, came forward, his neck bent, his head hanging, a handkerchief held to his ear as if he were suffering from a terrible toothache.

All these people wore the blue blouse over quaint and antique coats of a black or greenish cloth, Sunday clothes which they would only uncover in the streets of Havre. Their heads were covered by silk caps at high as towers, the emblem of supreme elegance in the small villages of Normandy.

Cesaire Horlaville closed the door, climbed up on his box, and snapped his whip.

The three horses awoke and, tossing their heads, shook their bells.

The driver then yelling "Get up!" as loud as he could, whipped up his horses. They shook themselves, and, with an effort, started off at a slow, halting gait. And behind them came the coach, rattling its shaky windows and iron springs, making a terrible clatter of hardware and glass, while the passengers were tossed hither and thither like so many rubber balls.

At first, all kept silent out of respect for the priest, that they might not shock him. Being of a loquacious and genial disposition, he started the conversation.

"Well, Maitre Caniveau," said he, "how are you getting along?"

The enormous farmer who, on account of his size, girth and stomach, felt a bond of sympathy for the representative of the Church, answered with a smile:

"Pretty well, Monsieur le cure, pretty well. And how are you?"

"Oh! I'm always well and healthy."

"And you, Maitre Poiret?" asked the abbe.

"Oh! I'd be all right only the colzas ain't a-goin' to give much this year, and times are so hard that they are the only things worth while raisin'."

"Well, what can you expect? Times are hard."

"Hub! I should say they were hard," sounded the rather virile voice of Rabot's big consort.

As she was from a neighboring village, the priest only knew her by name.

"Is that you, Blondel?" he said.

"Yes, I'm the one that married Rabot."

Rabot, slender, timid, and self-satisfied, bowed smilingly, bending his head forward as though to say: "Yes, I'm the Rabot whom Blondel married."

Suddenly Maitre Belhomme, still holding his handkerchief to his ear, began groaning in a pitiful fashion. He was going "Oh-oh-oh!" and stamping his foot in order to show his terrible suffering.

"You must have an awful toothache," said the priest.

The peasant stopped moaning for a minute and answered:

"No, Monsieur le cure, it is not the teeth. It's my ear-away down at the bottom of my ear."

"Well, what have you got in your ear? A lump of wax?"

"I don't know whether it's wax; but I know that it is a bug, a big bug, that crawled in while I was asleep in the haystack."

"A bug! Are you sure?"

"Am I sure? As sure as I am of heaven, Monsieur le cure! I can feel it gnawing at the bottom of my ear! It's eating my head for sure! It's eating my head! Oh-oh-oh!" And he began to stamp his foot again.

Great interest had been aroused among the spectators. Each one gave his bit of advice. Poiret claimed that it was a spider, the teacher, thought it might be a caterpillar. He had already seen such a thing once, at Campemuret, in Orne, where he had been for six years. In this case the caterpillar had gone through the head and out at the nose. But the man remained deaf in that ear ever after, the drum having been pierced.

"It's more likely to be a worm," said the priest.

Maitre Belhomme, his head resting against the door, for he had been the last one to enter, was still moaning.

"Oh—oh—oh! I think it must be an ant, a big ant—there it is biting again. Oh, Monsieur le cure, how it hurts! how it hurts!"

"Have you seen the doctor?" asked Caniveau.

"I should say not!"

"Why?"

The fear of the doctor seemed to cure Belhomme. He straightened up without, however, dropping his handkerchief.

"What! You have money for them, for those loafers? He would have come once, twice, three times, four times, five times! That means two five-franc pieces, two five-franc pieces, for sure. And what would he have done, the loafer, tell me, what would he have done? Can you tell me?"

Caniveau was laughing.

"No, I don't know. Where are you going?"

"I am going to Havre, to see Chambrelan."

"Who is Chambrelan?"

"The healer, of course."

"What healer?"

"The healer who cured my father."

"Your father?"

"Yes, the healer who cured my father years ago."

"What was the matter with your father?"

"A draught caught him in the back, so that he couldn't move hand or foot."

"Well, what did your friend Chambrelan do to him?"

"He kneaded his back with both hands as though he were making bread! And he was all right in a couple of hours!"

Belhomme thought that Chambrelan must also have used some charm, but he did not dare say so before the priest. Caniveau replied, laughing:

"Are you sure it isn't a rabbit that you have in your ear? He might have taken that hole for his home. Wait, I'll make him run away."

Whereupon Caniveau, making a megaphone of his hands, began to mimic the barking of hounds. He snapped, howled, growled, barked. And everybody in the carriage began to roar, even the schoolmaster, who, as a rule, never ever smiled.

However, as Belhomme seemed angry at their making fun of him, the priest changed the conversation and turning to Rabot's big wife, said:

"You have a large family, haven't you?"

"Oh, yes, Monsieur le cure—and it's a pretty hard matter to bring them up!"

Rabot agreed, nodding his head as though to say: "Oh, yes, it's a hard thing to bring up!"

"How many children?"

She replied authoritatively in a strong, clear voice:

"Sixteen children, Monsieur le cure, fifteen of them by my husband!"

And Rabot smiled broadly, nodding his head. He was responsible for fifteen, he alone, Rabot! His wife said so! Therefore, there could be no doubt about it. And he was proud!

And whose was the sixteenth? She didn't tell. It was doubtless the first. Perhaps everybody knew, for no one was surprised. Even Caniveau kept mum.

But Belhomme began to moan again:

"Oh-oh-oh! It's scratching about in the bottom of my ear! Oh, dear, oh, dear!"

The coach just then stopped at the Cafe Polyto. The priest said:

"If someone were to pour a little water into your ear, it might perhaps drive it out. Do you want to try?"

"Sure! I am willing."

And everybody got out in order to witness the operation. The priest asked for a bowl, a napkin and a glass of water, then he told the teacher to hold the patient's head over on one side, and, as soon as the liquid should have entered the ear, to turn his head over suddenly on the other side.

But Caniveau, who was already peering into Belhomme's ear to see if he couldn't discover the beast, shouted:

"Gosh! What a mess! You'll have to clear that out, old man. Your rabbit could never get through that; his feet would stick."

The priest in turn examined the passage and saw that it was too narrow and too congested for him to attempt to expel the animal. It was the teacher who cleared out this passage by means of a match and a bit of cloth. Then, in the midst of the general excitement, the priest poured into the passage half a glass of water, which trickled over the face through the hair and down the neck of the patient. Then the schoolmaster quickly twisted the head round over the bowl, as though he were trying to unscrew it. A couple of drops dripped into the white bowl. All the passengers rushed forward. No insect had come out.

However, Belhomme exclaimed: "I don't feel anything any more." The priest triumphantly exclaimed: "Certainly it has been drowned." Everybody was happy and got back into the coach.

But hardly had they started when Belhomme began to cry out again. The bug had aroused itself and had become furious. He even declared that it had now entered his head and was eating his brain. He was howling with such contortions that Poirat's wife, thinking him possessed by the devil, began to cry and to cross herself. Then, the pain abating a little, the sick man began to tell how it was running round in his ear. With his finger, he imitated the movements of the body, seeming to see it, to follow it with his eyes: "There it goes up again! Oh—oh—oh—what torture!"

Caniveau was getting impatient. "It's the water that is making the bug angry. It is probably more accustomed to wine."

Everybody laughed, and he continued: "When we get to the Cafe Bourbeux, give it some brandy, and it won't bother you any more, I wager."

But Belhomme could contain himself no longer; he began howling as though his soul were being torn from his body. The priest was obliged to hold his head for him. They asked Cesaire Horlaverse to stop at the nearest house. It was a farmhouse at the side of the road. Belhomme was carried into it and laid on the kitchen table in order to repeat the operation. Caniveau advised mixing brandy and water in order to benumb and perhaps kill the insect. But the priest preferred vinegar. They poured the liquid in drop by drop this time, that it might penetrate down to the bottom, and they left it several minutes in the organ that the beast had chosen for its home.

A bowl had once more been brought; Belhomme was turned over bodily by the priest and Caniveau, while the schoolmaster was tapping on the healthy ear in order to empty the other.

Cesaire Horlaverse himself, whip in hand, had come in to observe the proceedings.

Suddenly, at the bottom of the bowl appeared a little brown spot, no bigger than a tiny seed. However, it was moving. It was a flea! First, there were cries of astonishment and then shouts of laughter. A flea! Well, that was a good joke, a mighty good one! Caniveau was slapping his thigh, Cesaire Horlaverse snapped his whip, the priest laughed like a braying donkey, the teacher cackled as though he were sneezing, and the two women were giving little screams of joy, like the clucking of hens.

Belhomme had seated himself on the table and had taken the bowl between his knees; he was observing, with serious attention and a vengeful anger in his eye, the conquered insect that was twisting round in the water. He grunted, "You rotten little beast!" and he spat on it.

The driver, wild with joy, kept repeating: "A flea, a flea, ah! there you are, damned little flea, damned little flea, damned little flea!" Then having calmed down a little, he cried: "Well, back to the coach! We've lost enough time."

On the next page

Flea Hunt

by Bartolomeus Maton

(b. 1641 Leiden, South Holland, Netherlands, d. after 1684 Stockholm, Sweden)

Oil on oaken panel (26 x 21 cm)



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